Roll No. | | | | | | | |

ODD END SEMESTER EXAMINATION DECEMBER – 2025

Name of the Course: B. Tech (Civil Engineering) Semester: V

Name of the Paper: Air and Noise Pollution

Paper Code: TCE511

Time: 3 Hours Maximum Marks: 100

Note:-

(i) All questions are compulsory.

(ii) Answer any two sub questions among a, b & c in each main question

(iii) Total marks in each main question are twenty.

(iv) Each question carries 10 marks

Q1	**(20marks)**	
(a)	Define air pollutants and explain the primary and secondary sources of air pollution with suitable examples.	**CO1**
(b)	Explain the major pollutants emitted from stationary and mobile sources of combustion and their control measures.	
(c)	Compare and contrast London smog and Los Angeles (photochemical) smog in terms of composition, formation conditions, and effects.	
Q2	**(20 marks)**	
(a)	What are the methods used for particulate matter measurement? Discuss high-volume sampling and PM2.5/PM10 monitoring techniques.	**CO2**
(b)	List the National Ambient Air Quality Standards (NAAQS) as per the Central Pollution Control Board (CPCB), India. Explain how these standards are categorized for industrial, residential, and sensitive areas.	
(c)	Define Air Quality Index (AQI). Describe its computation method and the interpretation of AQI categories with their corresponding health impacts.	
Q3	**(20 marks)**	
(a)	Describe the mechanisms of gaseous pollutant removal by adsorption, absorption, and chemical reaction methods. Give suitable examples of pollutants and corresponding control devices for each method.	**CO3**
(b)	Define Indoor Air Quality (IAQ) and discuss the major indoor air pollutants, their sources, effects, and control strategies to maintain healthy indoor environments.	
(c)	Compare and contrast the operation and efficiency of wet collectors, fabric filters, and electrostatic precipitators (ESP) used for particulate pollution control.	
Q4	**(20 marks)**	
(a)	Explain the basic principles of acoustics and discuss how sound is characterized in terms of frequency, wavelength, and amplitude.	**CO4**
(b)	Describe the characteristics and examples of plane, point, and line sound sources. How does the nature of the source affect sound propagation?	
(c)	Explain the concept of psychoacoustics. How does human perception of sound influence the assessment of noise pollution?	
Q5	**(20 marks)**	
(a)	Discuss various noise control methods at the source, transmission path, and receiver level. Provide practical examples for each approach.	**CO5**
(b)	Explain the noise instrumentation and monitoring procedures, including the working principles of sound level meters and frequency analyzers.	
(c)	Describe the characteristics, sources, and effects of infrasound, ultrasound, impulsive sound, and sonic boom on humans and structures.	